INCH-POUND

MIL-PRF-1/1056E
22 July 1999
SUPERSEDING
MIL-E-1/1056D
23 March 1971

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
TYPE 12SP7D

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Magnetic deflection and focus.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

Parameter: Unit:	Ef V	Ec1 V dc	Ec2 V dc	Eb kV dc	Ehk V dc	Rg1 Meg	Alt ft
Maximum: Minimum:	6.9 5.7	0, -180 ---	770 ---	15 4	-125 ---	1/ ---	60,000 ---
Test conditions:	6.3	Adjust	250	9	---	---	---

See footnotes at end of table I.

GENERAL:

Qualification: Required.

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TABLE I. Testing and inspection.

Inspection	Method	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Qualification inspection								
Base material insulating quality	1216		---	---	---	---	---	---
Barometric pressure, reduced	1002	54 mmHg	---	---	---	---	---	---
Cathode illumination	5216		---	---	---	---	---	---
Direct-interelectrode capacitance	1331	Ck to all Cg1 to all Cg2 to all	--- --- ---	--- --- ---	Ck Cg1 Cg2	--- --- ---	8 10 10	pF pF pF
Pressure (implosion)	1141		---	---	---	---	---	---
Vibration (cathode-ray tubes)	5111		---	---	Width	---	2	mm
Conformation inspection part 1								
Bulb, screen, and faceplate quality	5106		---	---	---	---	---	---
Heater current	1301		---	---	If	540	660	mA
Voltage breakdown	5201		---	---	---	---	---	---
Voltage breakdown (magnetic types)	5201		---	---	---	---	---	---
Gas ratio	5206	2/	---	---	Gr	---	0.25	---
Neck straightness	---	3/	---	---	---	---	---	---
Cathode image	---	4/	---	---	---	---	20	%
Stray light emission (conventional types)	5216	Eb = 15,000 V dc; Ec2 = 770 V dc	---	---	---	---	---	---
Spot position (magnetic deflection)	5231	5/	---	---	---	---	20	mm
Zero-bias anode current (magnetic deflection)	5236		---	---	---	---	---	---
Grid cutoff voltage	5241		---	---	Eco	-27	-63	V dc
Grid No. 1 leakage current	5251		---	---	---	---	---	---
Grid No. 2 leakage current	5251		---	---	---	---	---	---
Aperture alignment	---	6/	---	---	Distance	---	8	mm

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 2								
Electrode current (grid No. 2 current)	5201	Ec1 = 90	---	---	Ic2	---	50	µA dc
Side terminal and base Alignment	5101	Pin No. 3	---	---	---	---	---	---
Base pin solder depth	1111		---	---	---	---	---	---
Neck and bulb alignment (magnetic types)	5101		---	---	---	---	.25	Inch
Face tilt	5101		---	---	---	---	---	---
Modulation	5223	Ib = 200 µA dc	---	---	ΔEc1	---	35	V dc
Screen	5221		---	---	---	---	---	---
Line width A (magnetic deflection)	5226	Ib = 200 µA dc	---	---	Width	---	0.38	mm
Line width C (magnetic deflection)	5226	Ib = 200 µA dc	---	---	Width	---	0.60	mm
Focusing ampere turns	5246	Ib = 200 µA dc; D = 3.25 inches	---	---	AT	450	550	---
Heater-cathode leakage Current	5251		---	---	---	---	---	---
Secureness of base, cap, or insert	1101		---	---	---	---	---	---
Permanence of marking	1105		---	---	---	---	---	---
Conformance inspection, part 3								
Life-test provisions	---	Group C; t = 500 hours (min); Eb = 13,200 V dc; Ec2 = 770 V dc; Ib = 60 µA dc	---	---	---	---	---	---
Life-test end points	---	Ib = 200 µA dc;						
		Line width A	---	---	Width	---	0.38	mm
		Line width B	---	---	Width	---	0.60	mm
		Modulation	---	---	ΔEc	---	35	V dc
		Heater-cathode leakage current	---	---	---	---	---	---
		Grid No. 1 leakage current	---	---	---	---	---	---
		Grid No. 2 leakage current	---	---	---	---	---	---
		Stray light emission (conventional types)	---	---	---	---	---	---

See footnotes at top of next page.

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TABLE I. Testing and inspection - Continued.

1/ When Ec2 is greater than 330 V dc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 V dc, Rg1 shall not exceed 1.5 Meg.

2/ This test to be performed at the conclusion of the holding period.

3/ The neck and base straightness shall be determined by the insertion of the tube neck in a cylinder 5 inches (127.00 mm) long and 1.5 inches (38.10 mm) maximum inside diameter. This cylinder shall move freely between the reference line and the base of the assembled tube.

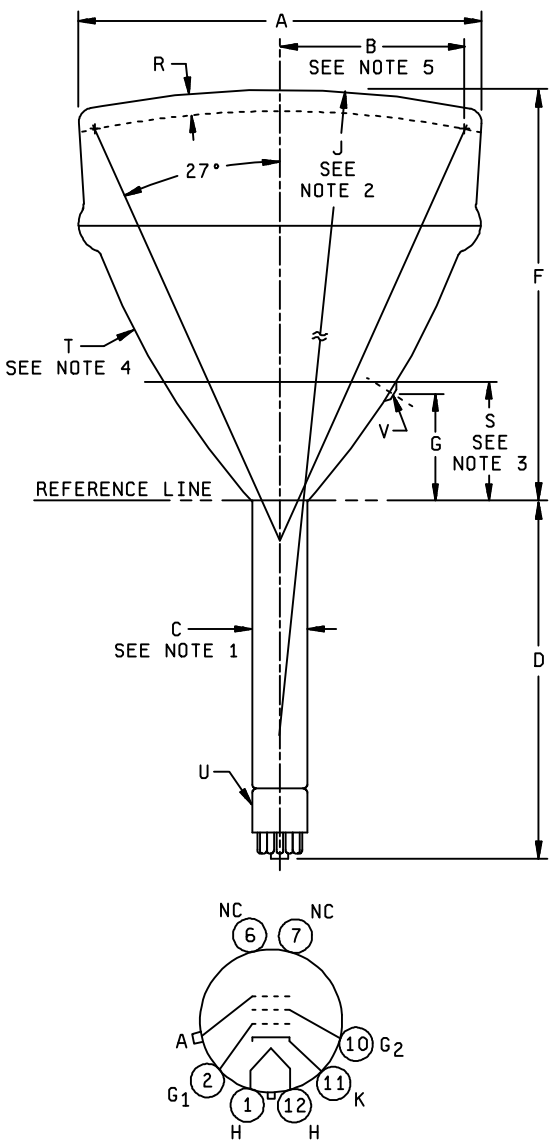
4/ The test conditions shall be Eb = 7 kV, Ec2 = 250, focus coil current = 0, Ec1 shall be varied until the focused image of the cathode is obtained. The percentage of dead area shall be that portion of the image which is black in respect to the general brilliant area.

5/ There shall be no displacement due to hand capacitance or external electrostatic field.

6/ The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. Ec1 shall not be held at zero for more than approximately 30 seconds to prevent damage to the screen.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	12.340	12.500	313.44	317.50
B	5.500	---	139.70	---
C	1.384	1.500	35.16	38.10
D	8.060	8.440	204.72	214.38
F	10.310	10.690	261.87	271.53
G	2.630	3.380	66.80	85.85
R	.300	.400	7.62	10.16
S	2.970	3.090	75.44	78.49
Conformance inspection, part 3 (periodic check)				
T	Bulb: J-99 1/2 A1, clear or filter glass			
U	Base: B7-51 or B5-57			
V	Contact: J1-21			
Reference dimensions (see note 6)				
J	40.000 R		1016.00 R	

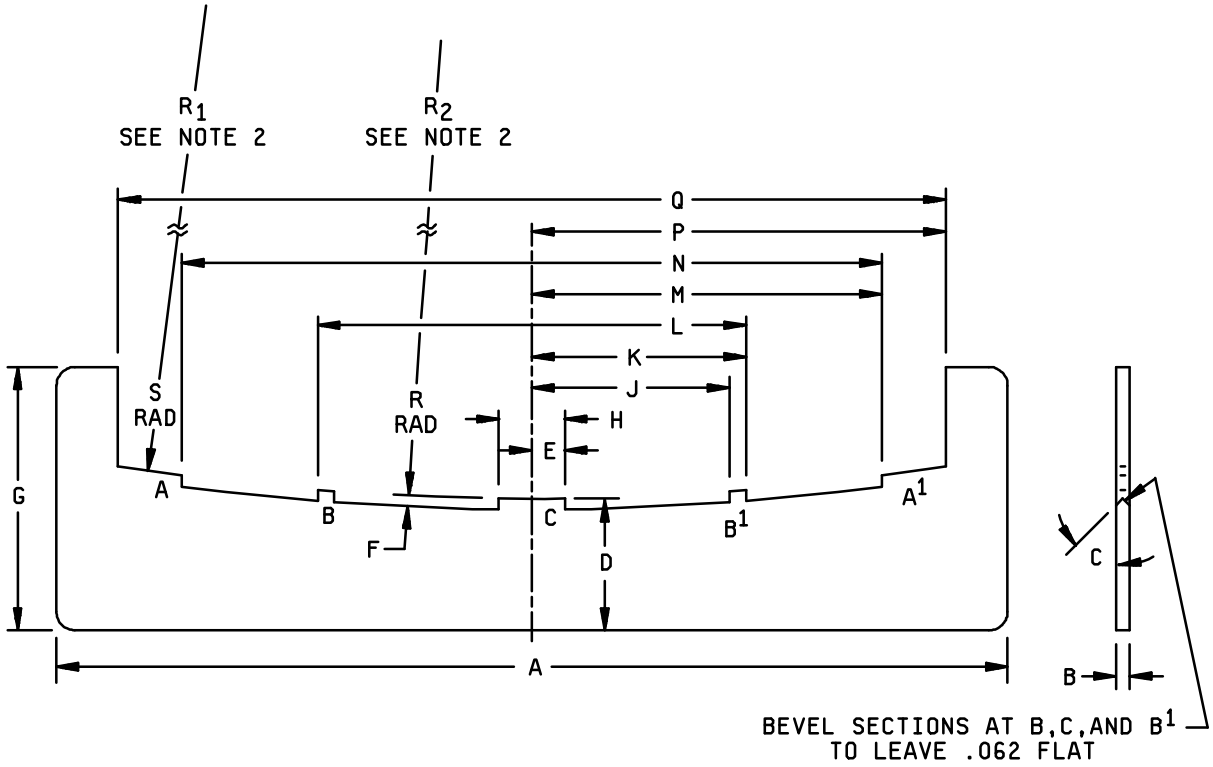


NOTES:

1. Dimension C applies to the entire length of neck below reference line. Reference line shall be determined by position where gauge 1.500 inches (38.10 mm) ± .003 inch (0.08 mm) I. D. and 2.000 inches (50.80 mm) long will rest on bulb cone.
2. To meet the requirements of faceplate gauge as shown on figure 2.
3. Line established by ring gauge of I. D. 7.188 inches (182.58 mm) ± .016 inch (0.41 mm).
4. To meet the requirements of bulb contour gauge as shown on figure 3.
5. Useful screen radius.
6. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 12SP7D.

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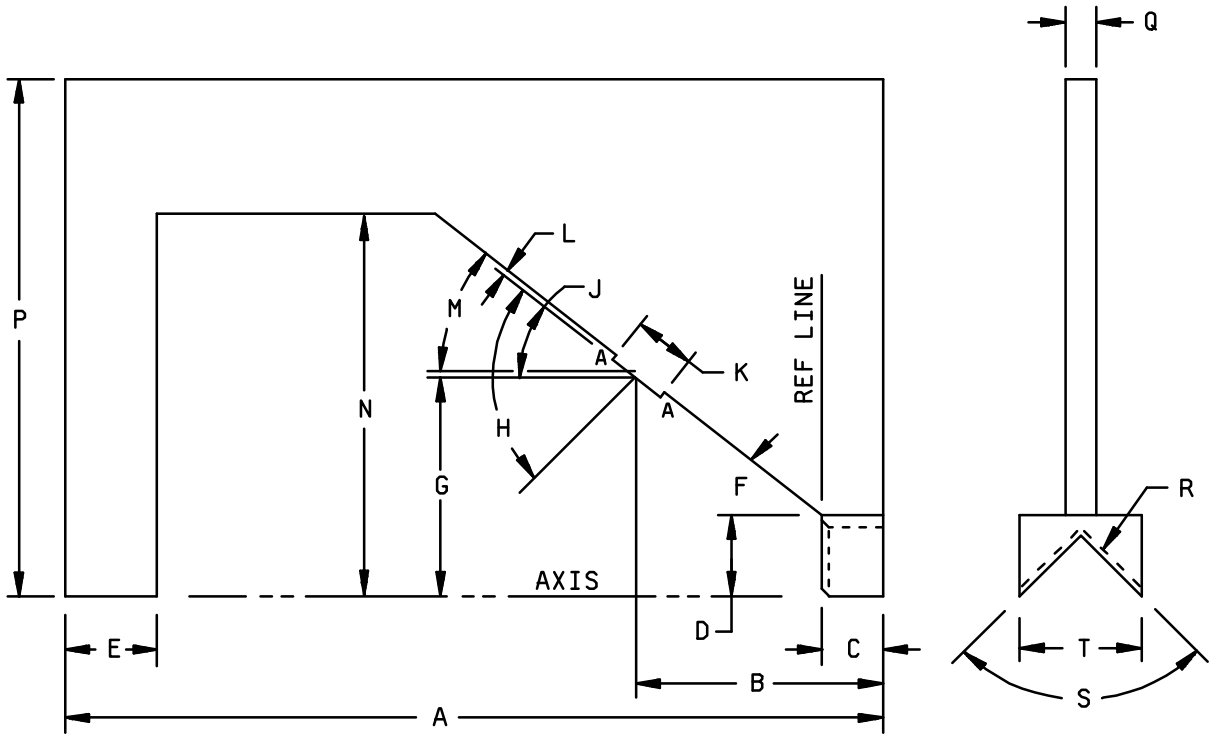
Dimensions									
Ltr	Inches		Millimeters		Ltr	Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
A	14.187	14.687	360.35	373.05	K	3.242	3.258	82.35	82.75
B	.115	.135	2.92	3.43	L	6.484	6.516	164.69	165.51
C	44°30'	45°30'	44°30'	45°30'	M	5.304	5.320	134.72	135.13
D	1.750	2.250	44.45	57.15	N	10.609	10.641	269.47	270.28
E	.492	.508	12.50	12.90	P	6.273	6.289	159.33	159.74
F	.052	.072	1.32	1.83	Q	12.546	12.578	318.67	319.48
G	3.750	4.250	95.25	107.95	R	39.875 R	40.125 R	1012.83 R	1019.18 R
H	.984	1.016	24.99	25.81	S	39.875 R	40.125 R	1012.83 R	1019.18 R
J	2.992	3.008	76.00	76.40					

NOTE

1. All dimensions are in inches.
2. R2 shall be equal to the absolute value of R1 ± .050 inch (1.27 mm) ± .002 inch (0.05 mm).
3. Instructions for use: Contour of face shall be checked with gauge at two orientations approximately 90 degrees. Reject if gap between bulb face and gauge points B, B^1, or C is greater than .100 inch (2.54 mm). Reject if bulb does not seat on steps A and A^1.

FIGURE 2. Faceplate gauge.

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Dimensions									
Ltr	Inches		Millimeters		Ltr	Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
A	13.370	13.380	339.60	339.85	L	.092	.096	2.34	2.44
B	4.029	4.033	102.34	102.44	M	35°	35° 30'	35°	35° 30'
C	.995	1.005	25.27	25.53	N	6.271	6.291	159.28	159.79
D	1.015	1.019	25.78	25.88	P	8.495	8.505	215.77	216.03
E	1.495	1.505	37.97	38.23	Q	.495	.505	12.57	12.83
F	37° 30'	38° 30'	37° 30'	38° 30'	R	.120 x 44° 30' CHAM	.130 x 45° 30' CHAM	3.05 x 44° 30' CHAM	3.30 x 45° 30' CHAM
G	3.592	3.596	91.24	91.34					
H	89° 30'	90° 30'	89° 30'	90° 30'	S	89° 30'	90° 30'	89° 30'	90° 30'
J	35° 30'	36° 30'	35° 30'	36° 30'	T	1.995	2.005	50.67	50.93
K	.995	1.005	25.27	25.53					

NOTE

1. All dimensions are in inches.
2. Unless otherwise specified, tolerances shall be ± .016 inch (4.06 mm) on fractions ± .005 inch (0.13 mm) on decimals, and ± .5° on angles.
3. Instructions for use: Reject if bulb does not rest on flat 'A-A' when bulb neck is seated in V block.

FIGURE 3. Bulb contour gauge.

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Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - AS, CG, MC
Air Force - 17, 19, 99

Preparing activity:
DLA - CC

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